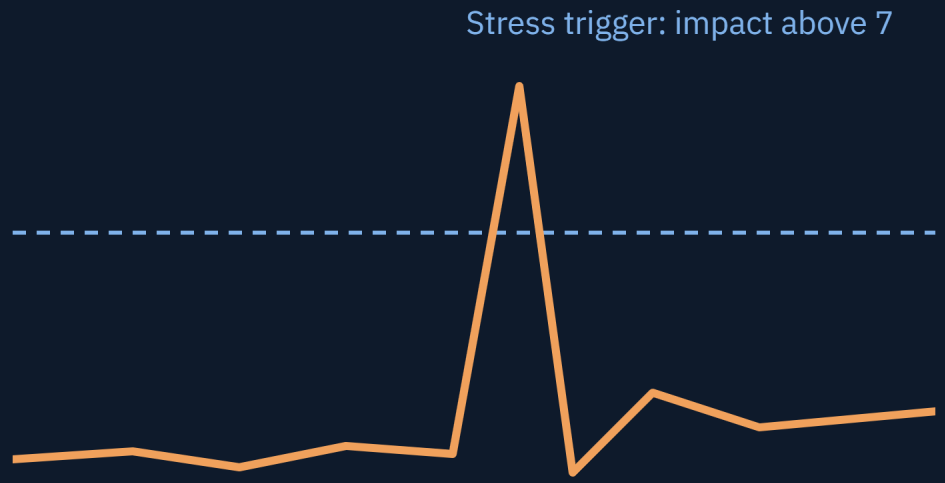
S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

ShockWire

Real-Time AI/NLP Financial Risk Engine

From headlines to risk signals to action.

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Risk arrives as text, faster than people can read it

A headline can move a portfolio in minutes. ShockWire reads news and social posts, scores each one, and hands the result to two modules that act on it.

1 TEXT IN

Two source types

Financial news and social posts, normalised into one format.

2 SIGNAL OUT

Three scores

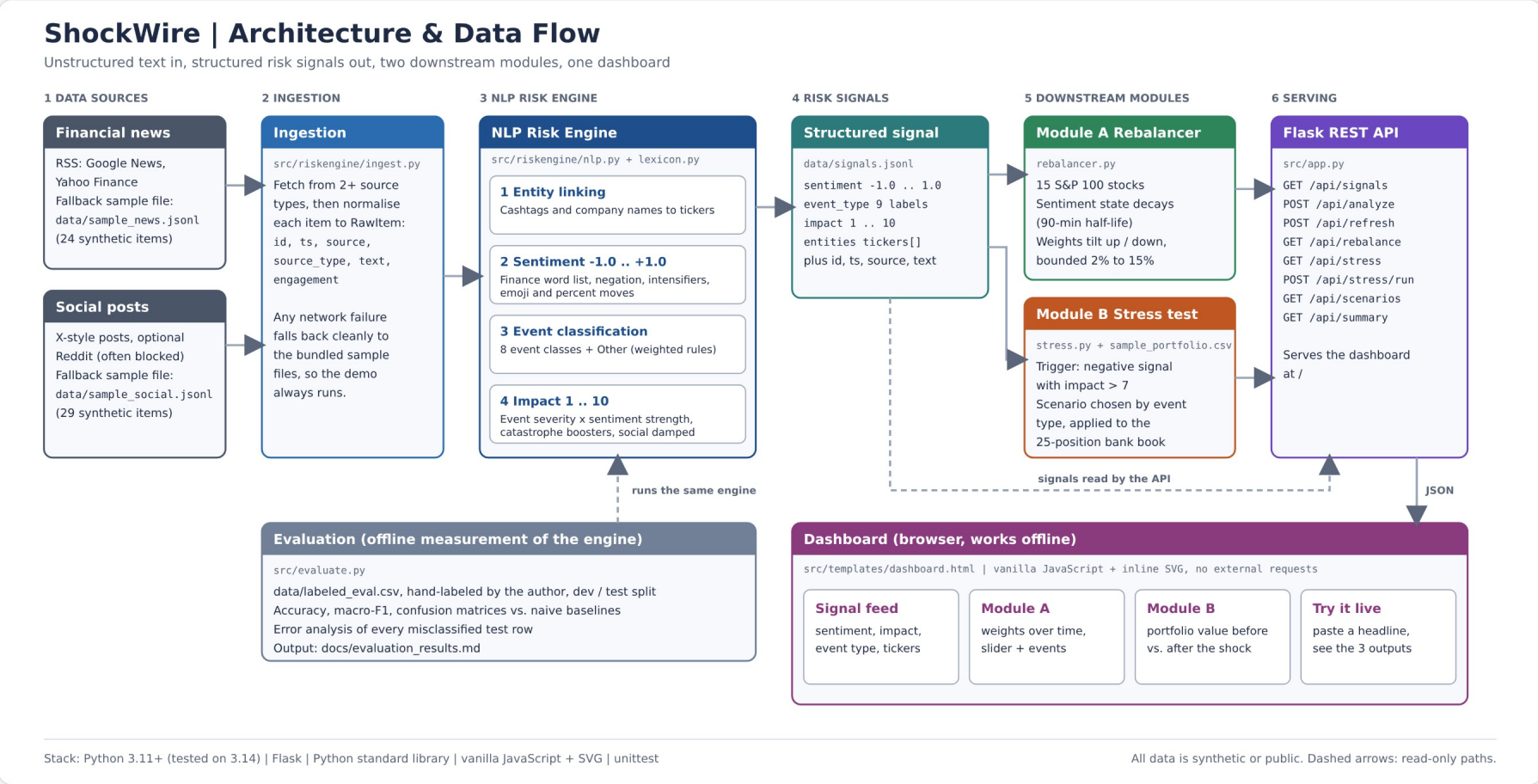
Sentiment from –1 to +1, event type, and impact from 1 to 10.

3 ACTION

Two modules

A rebalancer for a 15-stock index and a stress test for a 25-position bank book.

System design



Bundled sample files keep the demo running even when the network fails.

Built to be explainable and testable

NLP engine

A finance word list with negation, intensifiers, emoji and percent moves, plus weighted rules for 8 event classes. Every score traces back to words.

Impact score

Event severity times sentiment strength, plus catastrophe boosters. Social posts count less. A transparent heuristic, not a calibrated model.

Module A: rebalancer

Each stock keeps a sentiment state with a 90-minute half-life. Weights stay between 2% and 15% and always sum to 100%.

Module B: stress test

Triggers on negative news with impact above 7. Revalues loans, bonds, swaps, CDS, FX and equities position by position.

156 automated tests · runs offline · Flask is the only dependency

Key results

Stress loss on a $1,266.7M synthetic book



Derivative hedges gain while loans, bonds and equities lose.

Accuracy on 45 unseen headlines

Task	Engine	Baseline
Sentiment	62.2%	44.4%
Event type	62.2%	11.1%

Baseline is the majority class. Evaluated once, with no tuning afterwards: +17.8 and +51.1 points. A small set, so read it as an estimate.

Module A: weights follow sentiment through the sample day. One-way turnover is 93.7%, and transaction costs are not modelled.

Synthetic data. Stress shocks are illustrative, not regulator scenarios.

Domain impact

Speed and coverage

Screen far more text than a team can read, and see what matters first through the impact score.

From text to action

One signal drives a tactical response and a strategic one, with an audit trail in signals.jsonl.

Explainability

Every score traces to words and rules, which suits regulated risk work.

For wholesale banking, credit, market and counterparty exposure can be linked to an event the moment it appears in the news.

Limits and next steps

What it cannot do yet

- Event type is 98% accurate on the 103 headlines used during development, but 62% on 45 new ones. I report the 62%.
- Sarcasm, double negatives and informal posts are often missed.
- The demo day is scripted synthetic data, and the stress shocks are illustrative.
- The social feed is simulated, because Reddit blocked automated access.

Next steps

- Put a finance language model behind the same interface and compare it on the same unseen set.
- Calibrate impact against realised spread and volatility moves.
- Add regulator scenario libraries and correlations to the stress test.
- Add streaming ingestion and a human review queue for triggers.

Thank you · github.com/Medhanshug99/iiitranchi-medhansh-poojari-hackathon